

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

**ENTERPRISE WATER DISTRIBUTION AND FLEET MANAGEMENT
PLATFORM FOR OPTIMIZED OPERATIONS AND COST EFFICIENCY**

In partial fulfillment for the requirements in capstone 1

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Water distribution is a critical service essential for sustaining communities and supporting various industrial operations. In highly urbanized areas like Iloilo City, ensuring a consistent and reliable supply of water requires a well-coordinated logistics framework. However, many organizations today face significant challenges that hinder operational efficiency and increase costs. A primary concern is the lack of optimized delivery routes and the underutilization of fleet resources, which often results in delays, excessive fuel consumption, and an inability to meet the demands of all service areas on time.

Despite the growing complexity of these operations, many water distribution systems still rely on manual tracking and traditional scheduling methods. This lack of automation makes it difficult for managers to respond effectively to operational issues through timely monitoring and recorded system data, such as vehicle breakdowns, traffic delays, or sudden spikes in demand. This is particularly evident in developing residential and commercial zones, such as Hibao-an along the Circumferential Road, where rapid urbanization has led to increased demand and logistical bottlenecks. Without access to centralized monitoring and structured operational data in these high-traffic corridors, decision-making remains reactive, preventing organizations from achieving optimal service delivery.

As the scale of operations grows, the need for a more sophisticated management system becomes evident. A centralized platform that integrates fleet management with water distribution monitoring offers a strategic solution to these challenges. By leveraging technology to optimize routing and track resource allocation, organizations can significantly lower operational costs while improving service reliability. This study aims to develop a comprehensive system that addresses these systemic gaps, providing a more organized, cost-efficient, and reliable approach to water distribution and fleet management for the Hibao-an area and its surrounding communities.

1.2 Statement of the Problem

While water distribution is essential for urbanized areas like Iloilo City, the current logistical and operational frameworks face significant unresolved challenges. This study seeks to address the following problems:

1. **Inefficient Manual Scheduling:** Existing systems rely on traditional, manual tracking methods that are prone to human error and slow response times.
 2. **Lack of Optimized Routing:** The absence of automated route planning leads to excessive fuel consumption, vehicle wear-and-tear, and significant delivery delays.
 3. **Limited Real-Time Visibility:** Managers lack the tools to monitor fleet locations and resource allocation in real-time, making it difficult to respond to traffic bottlenecks or vehicle breakdowns.
 4. **Inadequate Response to Rapid Urbanization:** Current distribution methods struggle to scale with the increasing water demands of developing zones.
 5. **High Operational Overheads:** The underutilization of fleet resources and uncoordinated logistics result in unnecessarily high costs for water distribution providers.
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1.3 OBJECTIVE OF THE STUDY

1.3.1 GENERAL OBJECTIVE

The general objective of this study is to design and develop a centralized fleet management and water distribution system that utilizes **system monitoring** and automated routing to optimize logistical efficiency, enhance service reliability, and reduce operational costs.

1.3.2 Specific Objectives

To achieve this goal, the study aims to:

1. Assess the logistical requirements and existing limitations of manual water distribution methods in Ilo-Ilo City areas.
2. Design a secure and intuitive user interface for both fleet managers and delivery personnel to monitor active operations.

3. Develop an automated routing algorithm to optimize delivery paths and minimize fuel consumption.
 4. Implement system-based monitoring and resource tracking features to provide visibility into fleet operations, delivery progress, and cost-related data.to provide full visibility into fleet locations and delivery status.
 5. Construct a centralized administrative module for the management of vehicle schedules, driver assignments, and demand forecasting.
 6. Manage operational data, including delivery logs and fleet maintenance records, using a secure and structured database.
 7. Evaluate the system's effectiveness in reducing delivery delays, improving resource utilization, and maintaining performance under peak demand.
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1.4 Scope and Delimitation

The primary focus of this study is the development of a Web System designed for the centralized management of fleet operations and water distribution logistics. The system is specifically engineered for internal use by authorized personnel and the business owner to streamline high-traffic distribution corridors. The system focuses on **cost monitoring, operational tracking, and route optimization** using system-encoded data and trip logs. It does not utilize GPS-based real-time tracking; instead, monitoring is based on recorded inputs, delivery reports, and administrative updates within the system.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Water Distribution Owners** – by providing a high-level overview of operations through analytics and reports, allowing for better cost control and informed decision-making.

- **Authorized Personnel (Administrators)** – by streamlining the complexity of scheduling and driver management, reducing manual errors, and enabling effective oversight of the delivery process.
- **Fleet Drivers** – by providing organized schedules and optimized routes that reduce travel time, fuel consumption, and the stress associated with uncoordinated logistics.
- **Iloilo City Community** – by indirectly improving the reliability, consistency, and timeliness of water delivery services through more efficient resource allocation in rapidly urbanizing city zones.

1.6 Definition of Terms

- **Automated Routing Algorithm** – a computational procedure used to determine the most efficient path for delivery vehicles based on distance, traffic conditions, and delivery priorities.
 - **Centralized Administrative Module** – a primary control interface within the Web application where the owner and authorized personnel manage all system data, including driver profiles and fleet status.
 - **Database Management System (DBMS)** – the software used to store, retrieve, and manage operational data such as delivery logs, vehicle maintenance records, and user credentials.
 - **Fleet Management** – the administrative approach used to coordinate a company's vehicles to ensure they are used efficiently, safely, and maintained properly.
 - **Logistical Efficiency** – the measurement of how effectively resources, such as fuel and time, are utilized during the distribution of water to minimize waste and cost.
 - **Operational Overheads** – the ongoing expenses related to the daily running of the business, specifically focusing on fuel costs, vehicle repairs, and labor hours in this study.
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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Miller and Thompson (2021) discussed the application of automated routing algorithms in urban logistics platforms. Their study revealed that optimization models improved delivery accuracy by 25% by analyzing traffic patterns and historical fuel consumption data. Similarly, Chen et al. (2020) emphasized that fleet monitoring systems significantly enhance operational transparency and stakeholder trust in resource distribution applications. Furthermore, Wang (2022) noted that centralized Web-based management systems are preferred for administrative oversight in small-to-medium enterprises due to their robust data handling capabilities.

2.2 Local Studies

According to **Dela Cruz and Santos (2022)**, logistics-based systems developed in Philippine academic settings improve process automation and data accuracy, particularly in the context of local delivery services. Their study on intelligent monitoring applications highlights the importance of localized data in overcoming unique regional infrastructure challenges. **Reyes (2021)** also emphasized that integrating data-driven decision-making into water utility management enhances system reliability and reduces the response time to sudden spikes in community demand, which is critical in rapidly urbanizing Philippine cities like Iloilo.

2.3 Related Literature

Logistics management literature explains that route optimization relies on advanced algorithmic modeling to determine the most efficient and cost-effective delivery paths (Toth & Vigo, 2021). In the realm of software engineering, Pressman (2019) emphasizes the necessity of structured development methodologies to guarantee system reliability, particularly when integrating

real-time data analytics and cloud-based tracking APIs. Furthermore, research onunderscores the critical role of operational visibility and fuel consumption monitoring in minimizing overhead costs and preventing the underutilization of corporate assets (Goldsby et al., 2018).

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies demonstrate that automated routing and real-time monitoring significantly improve logistical intelligence and operational decision-making. However, much of the existing research focuses on large-scale commercial logistics or consumer-facing delivery apps, with limited focus on specialized, internal Web systems designed for water distribution management in rapidly urbanizing local contexts like Iloilo City. This study addresses this gap by integrating monitoring-based tracking and cost analysis and automated scheduling into a centralized administrative platform. By providing the owner and authorized personnel with direct oversight of distribution costs and fleet efficiency, the study aligns with current technological trends while providing a practical solution to localized infrastructure challenges.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a **developmental research design**, focusing on the design, development, and evaluation of the proposed **Enterprise Water Distribution and Fleet Management Platform**. This design follows a systematic process to create a functional software solution that addresses the logistical inefficiencies identified in Iloilo City.

3.2 System Development Methodology

The Waterfall Model will be used, consisting of the following phases:

1. Requirements Analysis
 2. System Design
 3. Implementation
 4. Testing
 5. Deployment (Prototype)
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3.3 System Architecture

A dedicated, responsive portal where users perform administrative tasks, monitor fleet movements in real-time, and manage cost oversight. By moving to a web-based model, the system ensures seamless access across the internal network without the need for local software installations.

3.4 Tools and Technologies Used

- **Frontend:** Html,css and JavaScript
 - **Backend:.**Python flask
 - **Database:** Firebase
 - **Development Tools:** Visual Studio Code
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3.5 Data Gathering Techniques

- Interview with the Client
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3.6 Testing Procedures

- User Acceptance Testing (UAT)
 - Unit Testing
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3.7 Evaluation Criteria:

The system will be evaluated based on the following criteria:

- Functionality
- Usability
- Security

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

- User (Admin) → Water Distribution System → Database
- User submits data
- System processes data

- Results are stored and retrieved
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3.11 Use Case Diagram

Actors:

- Administrator (Main Client)
- System (Machine Learning Engine)
- Fleet Staff (Drivers)

Use Cases:

- Manage Fleet Vehicles (Admin)
 - Input Water Orders (Admin)
 - Generate ML Cost Predictions (System)
 - Optimize Delivery Routes (System)
 - Monitor Fuel Expenses (Admin)
 - Manage User Accounts (Admin)
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End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging

technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.